Decanters

- Separation of two liquid phases, immiscible or partially miscible liquids, is a common requirement in the process industries.
- The simplest form of equipment used to separate liquid phases is the gravity settling tank, the decanter (settlers)
- The working principle of decanters is based on sedimentation or settling
- Immiscible is the property where two substances are not capable of combining to form

Examples

Oil and water,

Kerosene and Water.

Gasoline (Petrol) and Water.

Corn Syrup and Vegetable Oil.

Wax and Water.



Continuous Gravity

- A continuous gravity decanter is used for the separation of two immiscible liquids when the difference of densities of the two liquid is large



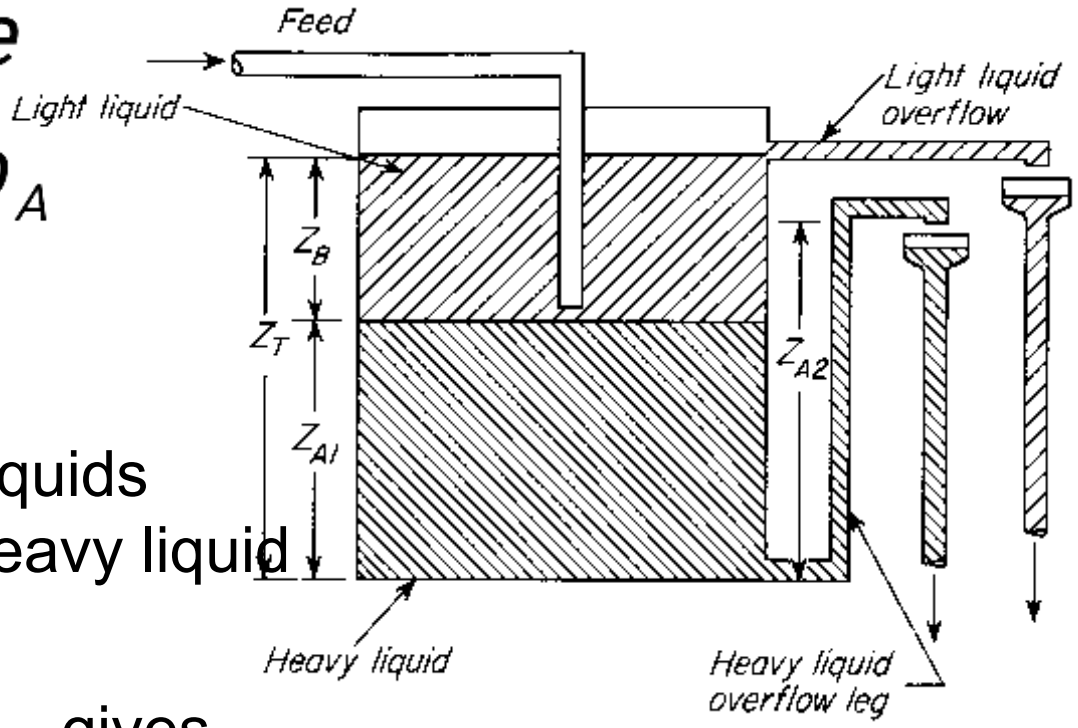
- The feed mixture enters at one end of the separator, the two liquids flow slowly through the vessel separate into two layers , and discharge through over flow lines at the other end of the separator

- The overflow lines are so large that the frictional resistance to the flow the liquids is negligible
- The performance of the decanter is analyzed by the principle of Fluid static
- Let the densities of the heavy and light liquids be ρa and ρband respectively.
- Let the interface between two liquids is at a height Z_{A1} from the bottom of the vessel.
- The total depth of the liquid of the vessel is Z_T and the depth of the layer of the light liquid in the separator is Z_{A1}
- The overflow of the light liquid is at a height Z_{A2} and the heavy liquid is at a height Z_T from the bottom of the vessel.
- Assume that the overflow lines and the vessel itself are open to the atmosphere through a vent line.

Continuos Gravity Decanter

Hydrostatic Balance

$$Z_B \rho_b + Z_{A1} \rho_A = Z_{A2} \rho_A$$



Where,

Z_B=depth of the light liquid

Z_{A1}=interface between two liquids

Z_{A2}=height of the overflow heavy liquid

Z_T=total depth of the liquid

Solving the equation for Z_{A1} gives

$$Z_{A1} = Z_{A2} - Z_B \frac{\rho_B}{\rho_A} = Z_{A2} - (Z_T - Z_{A1}) \frac{\rho_B}{\rho_A}$$

$$Z_{A1} = \frac{Z_{A2} - Z_T\left(\rho_B / \rho_A\right)}{1 - \rho_B / \rho_A}$$

When $\rho_B \approx \rho_A$ interface location is very sensitive to height of heavy liquid overflow leg. This leg is often has adjustable height to give the best separation.

USE OF DECANTER IN INDUSTRY

Sugar industry

Wine industry(Breweries)

Pharmaceutical industry

Food industry

Centrifugal decanters

When the density difference between two immiscible liquids is small gravitation forces may be too weak to separate them in a reasonable time. In this case we can use centrifugal forces to amplify the forces exerted on the liquids.

Centrifugal separations are important in many food industries such a breweries, vegetable oil processing, fruit juice processing. They are also used to separate emulsions into their components.

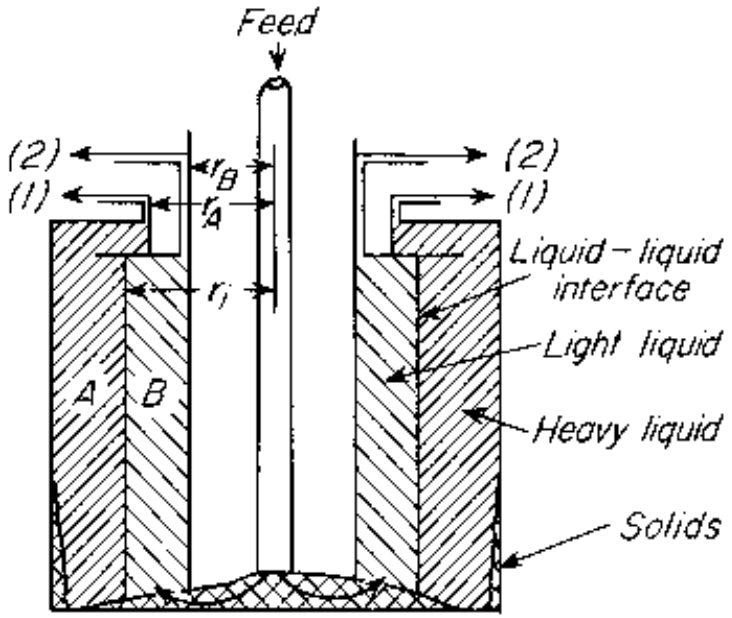
- A centrifuge is a device, which employs a high rotational speed to separate components of different densities.
- This becomes relevant in the majority of industrial jobs where solids, liquids and gases are merged into a single mixture and the separation of these different phases is necessary.
- A decanter centrifuge separates solid materials from liquids in slurry and therefore plays an important role in wastewater treatment, chemical, oil and food processing industries.
- There are several factors that affect the performance of a decanter centrifuge and some design heuristics to be followed which are dependent upon given applications.

Continuous Centrifugal Decanter

Consequences

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- $\rho_A \rho_B$ within 3% r_i unstable
- r_B constant r_A increased r_i shifted toward bowl wall
- In commercial units r_A and r_B are usually adjustable



TYPES OF DECANTER CENTRIFUGES

- **VERTICAL DECANTER CENTRIFUGE**
- **HORIZONTAL DECANTER CENTRIFUGE**



WORKING PRINCIPLE OF A DECANTER CENTRIFUGE

Step 1: The mixture slurry or two liquids is allowed into the centrifuge through a connecting pipe and onto a conveyor.



Step 2: The conveyor ushers the slurry mixture through a nozzle into the bowl area.

Step 3: The bowl rotates at high speeds to induce centrifugal forces.

WORKING PRINCIPLE OF A DECANTER CENTRIFUGE

Step 4: High speed rotation separates the solid material from the liquid in a matter of seconds.



Step 5: The conveyor delivers the solid material upwards, where it is discharged through a nozzle.

Step 6: With the solid material removed, the purified liquid is released from a separate output.

- Sedimentation of particles occurs in centrifugal field
- Flow is upwards at a particular rate which determines residence time in device
- Separation happens if sedimentation velocity is high enough for particle to reach side of bowl within residence time
- Large particles have higher settling velocities than small particles
- Both large and small are still particles, have small Reynolds no.s (<1) and obey Stokes' Law

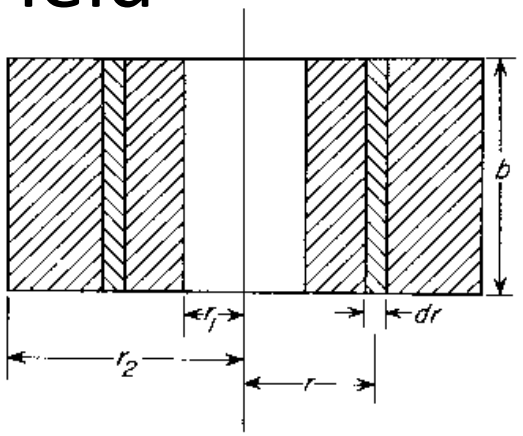


Hydrostatic Equilibrium in a Centrifugal Field

$$F_c = mr\omega^2 = mr\left(\frac{2\pi N}{60}\right)^2$$

$$F_g = mg$$

$$\frac{F_c}{F_g} = \frac{r}{g}\left(\frac{2\pi N}{60}\right)^2$$



Typically N≈1000 and r≈1m. Fc/Fg≈110.

Hydrostatic Equilibrium in Centrifugal Field

To calculate the pressure drop over any ring of rotating liquid

$$Force\ on\ element\ dr\ at\ r$$

$$dF = \omega^2 r\, dm$$

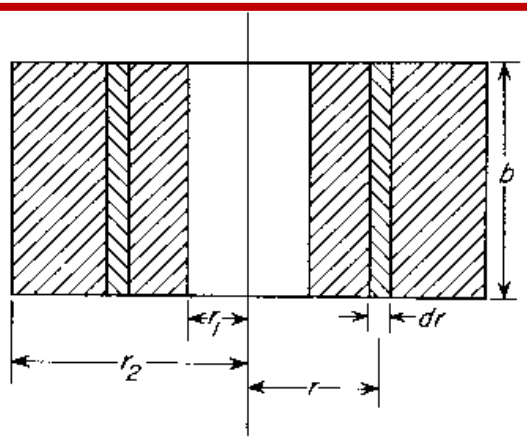
$$dm = 2\pi\rho r b\, dr$$

dF=centrifugal force
dm= mass of the liquid in the element
ω = angular velocity in rad/s

dm =volume x density

Eliminating dm

$$dF = 2\pi\rho b\omega^2 r^2\, dr$$

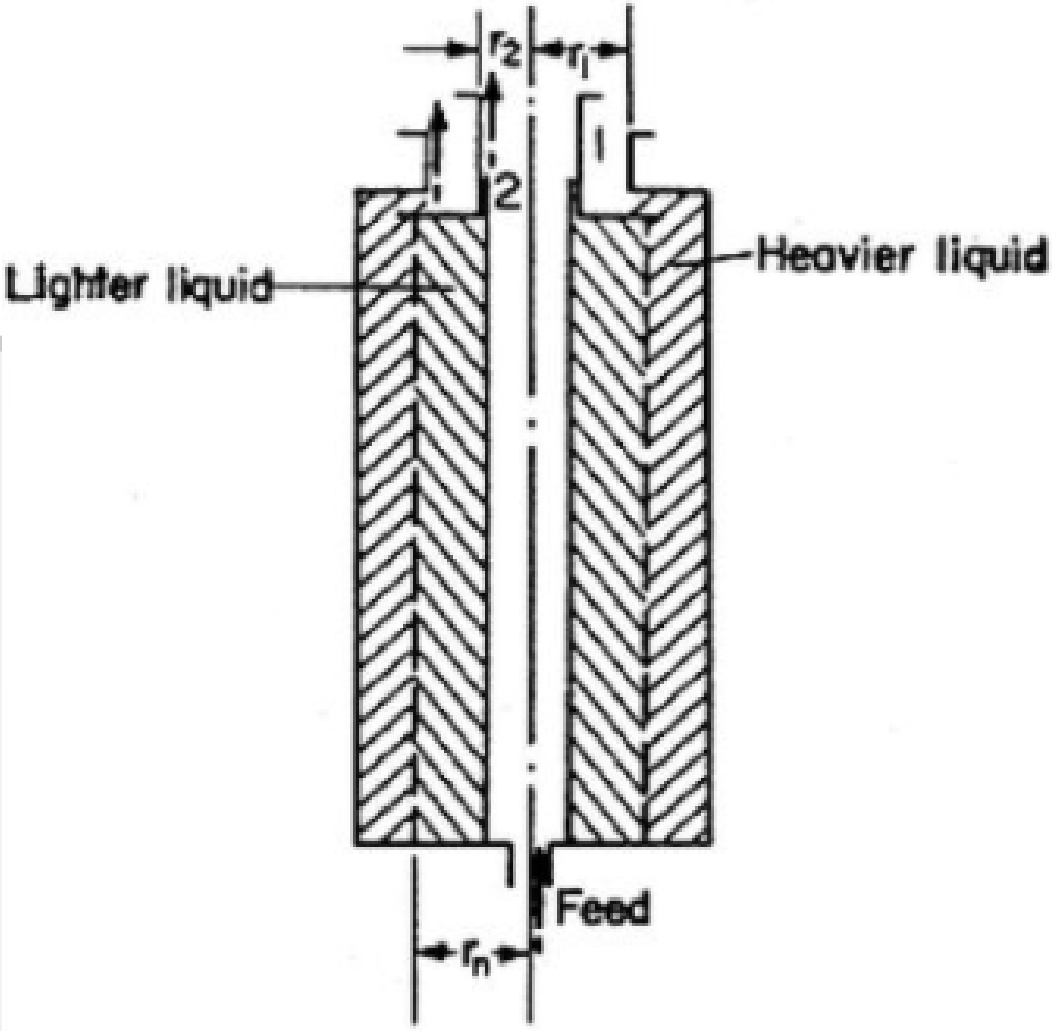


dp=force exerted in the element divided by the area of the ring

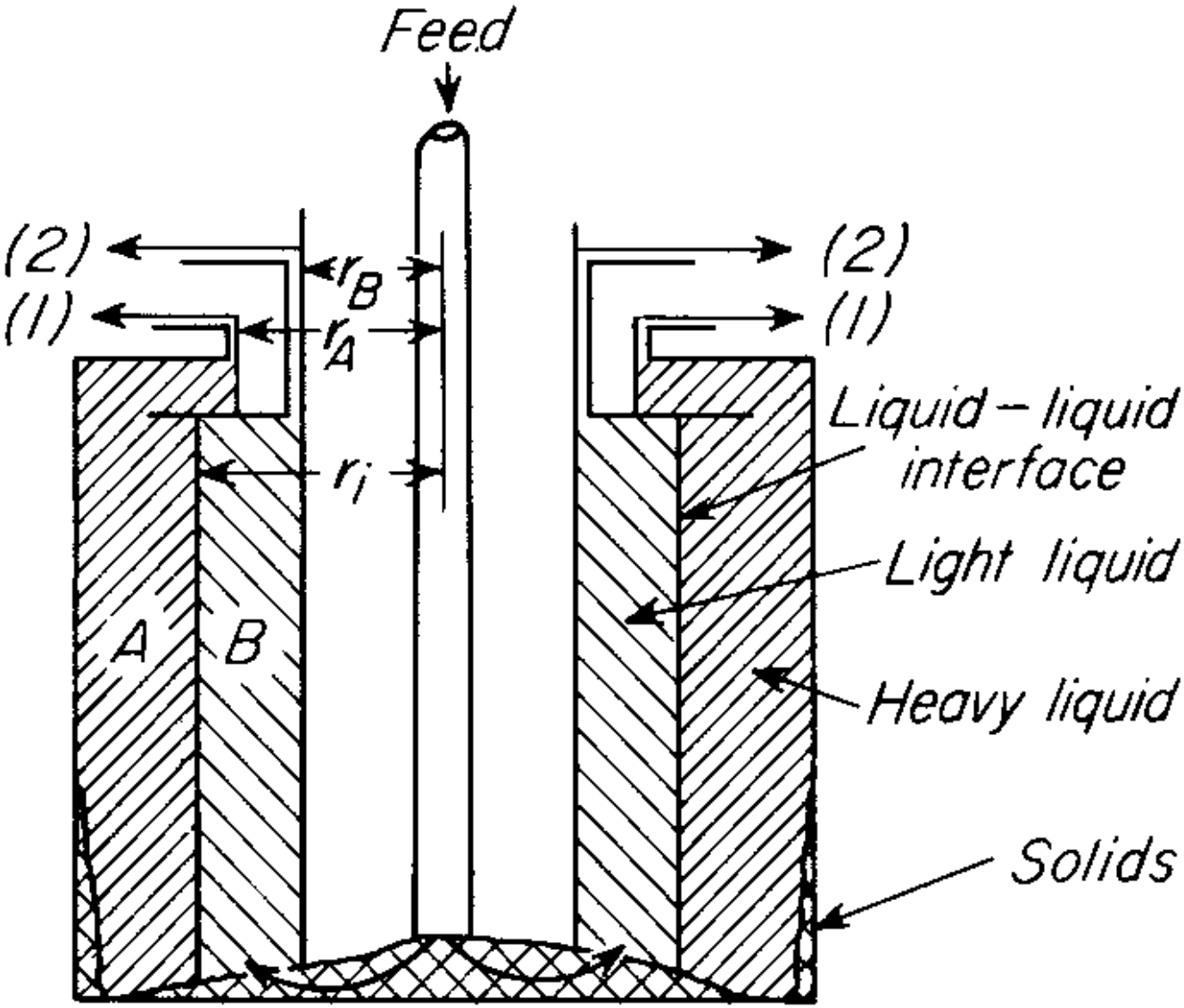
$$dP = \frac{dF}{2\pi r b} = \omega^2 \rho\, r\, dr$$

$$P_2 - P_1 = \frac{\omega^2 \rho}{2}\left(r_2^2 - r_1^2\right)$$ → Equation 1

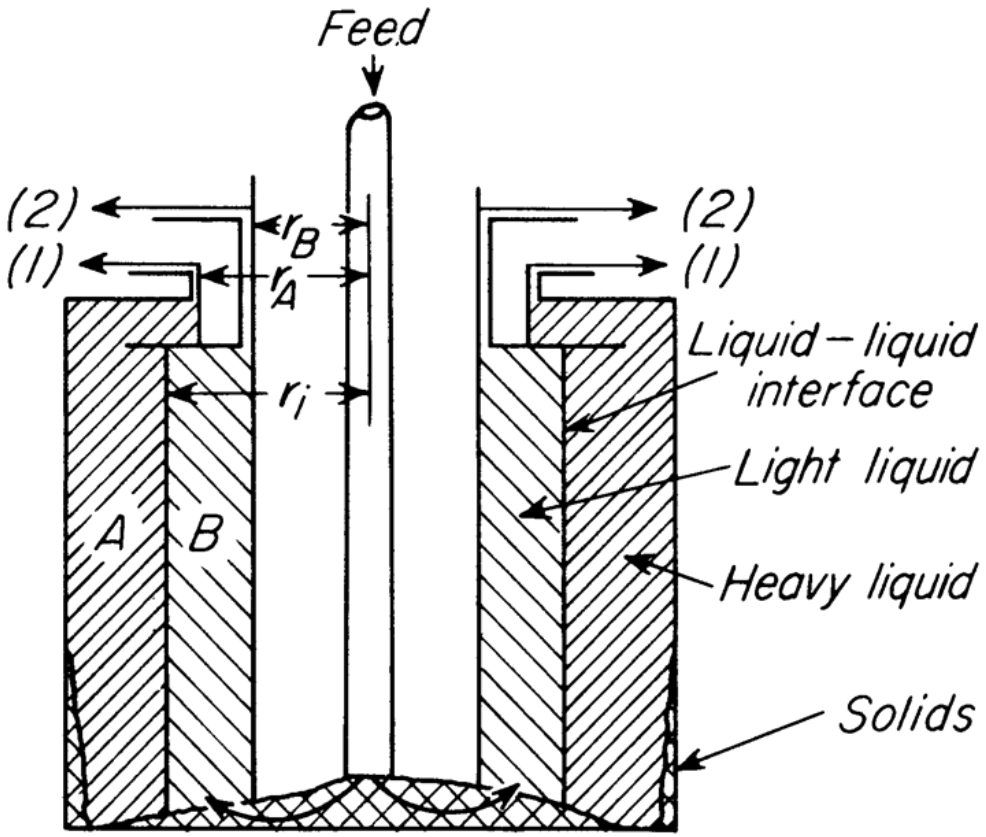
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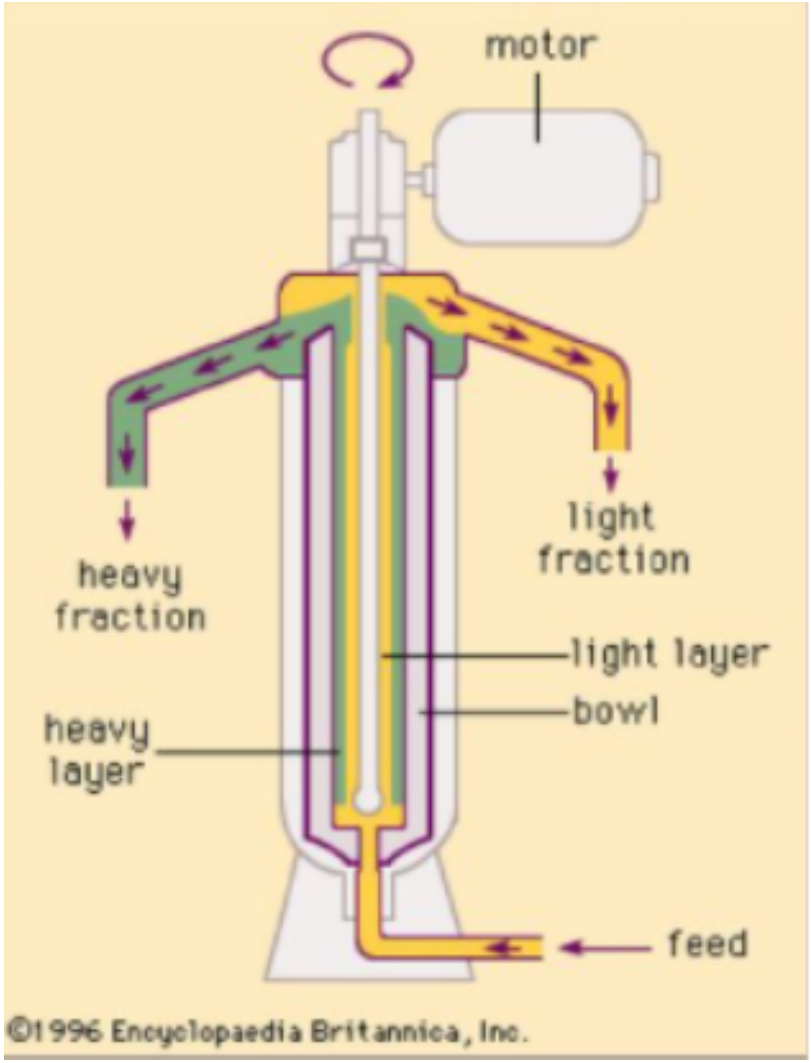


Continuous Centrifugal Decanter



- When the bowl with the two liquids is rotated at high speed. Heavy liquid attaches to the wall and the lighter liquid nearer the axis. Let A be the heavy liquid and B lighter liquid. A cylindrical interface r_i separated the two layer. It is referred as neutral zone.
- In operation of the machine the feed is admitted near the bottom of the bowl. Light liquid discharge through port 2 and heavy liquid discharge through port 1 .The liquid liquid interface is established by a hydrostatic balance.
- Let the heavy liquid of density ρ_A overflow the dam at radius r_A and light liquid of density ρ_B leaves through the ports at radius r_B
- At equilibrium the pressure difference in the light liquid between r_B and r_i must be equal to that in the heavy liquid between r_B and r_i.
- The principle exactly same as in a gravity decanter.


Feed
(2)
(1)
r_B
r_A
r_i
(2)
(1)
Liquid – liquid interface
Light liquid
Heavy liquid
Solids
A
B


motor
heavy fraction
light fraction
light layer
bowl
heavy layer
feed
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Continuous Centrifugal Decanter

Hence at equilibrium

$$P_i - P_A = P_i - P_B$$

Pi= pressure at liquid liquid interface

PB= pressure at the free surface of light liquid at rB

PA= pressure at the free surface of heavy liquid at r

From equation 1

$$P_i - P_A = \frac{\omega^2 \rho_A \left(r_i^2 - r_A^2\right)}{2}$$

$$P_i - P_B = \frac{\omega^2 \rho_B \left(r_i^2 - r_B^2\right)}{2}$$

$$\rho_A\left(r_i^2 - r_A^2\right) = \rho_B\left(r_i^2 - r_B^2\right)$$

Solving for ri

$$r_i = \sqrt{\frac{r_A^2 - \left(\rho_B / \rho_A\right) r_B^2}{1 - \rho_B / \rho_A}}$$

This equation is similar to the equation derived for gravity decanter.

The above equation shows that the radius of the neutral zone is sensitive to the density ratio.

Note:The difference between the two densities should not be less than approximately 3 percent for stable operation.

If rB is constant and rA is increased ri will be shifted toward bowl wall and the reverse is true.

In commercial units rA and rB are usually adjustable so that (if the separation in zone B is more difficult than that in zone A, zone B should be large and zone A should be small).